CAM/RF

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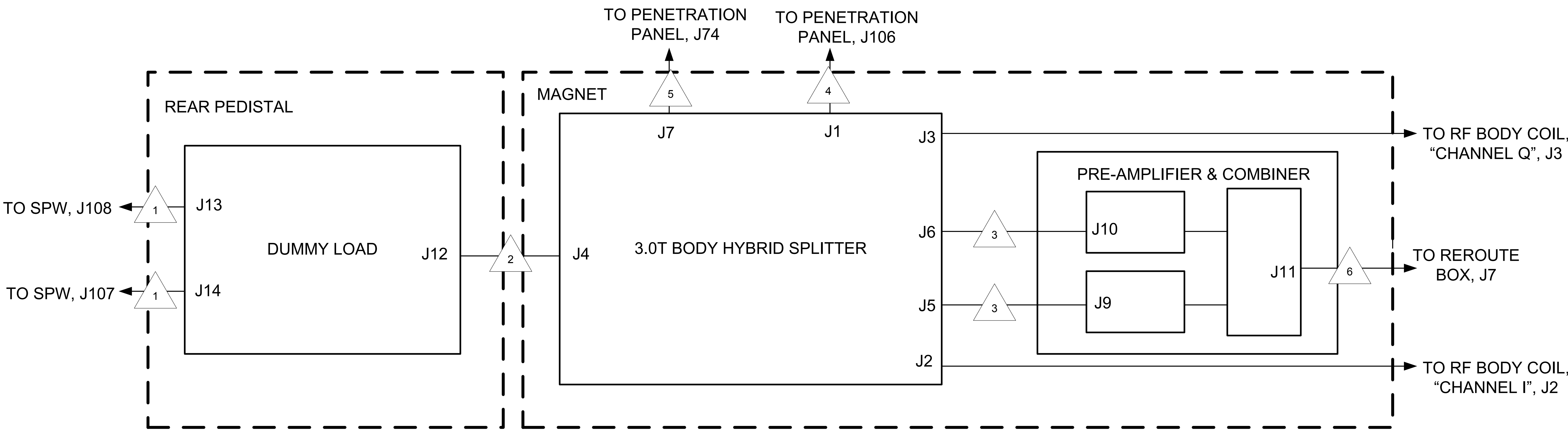
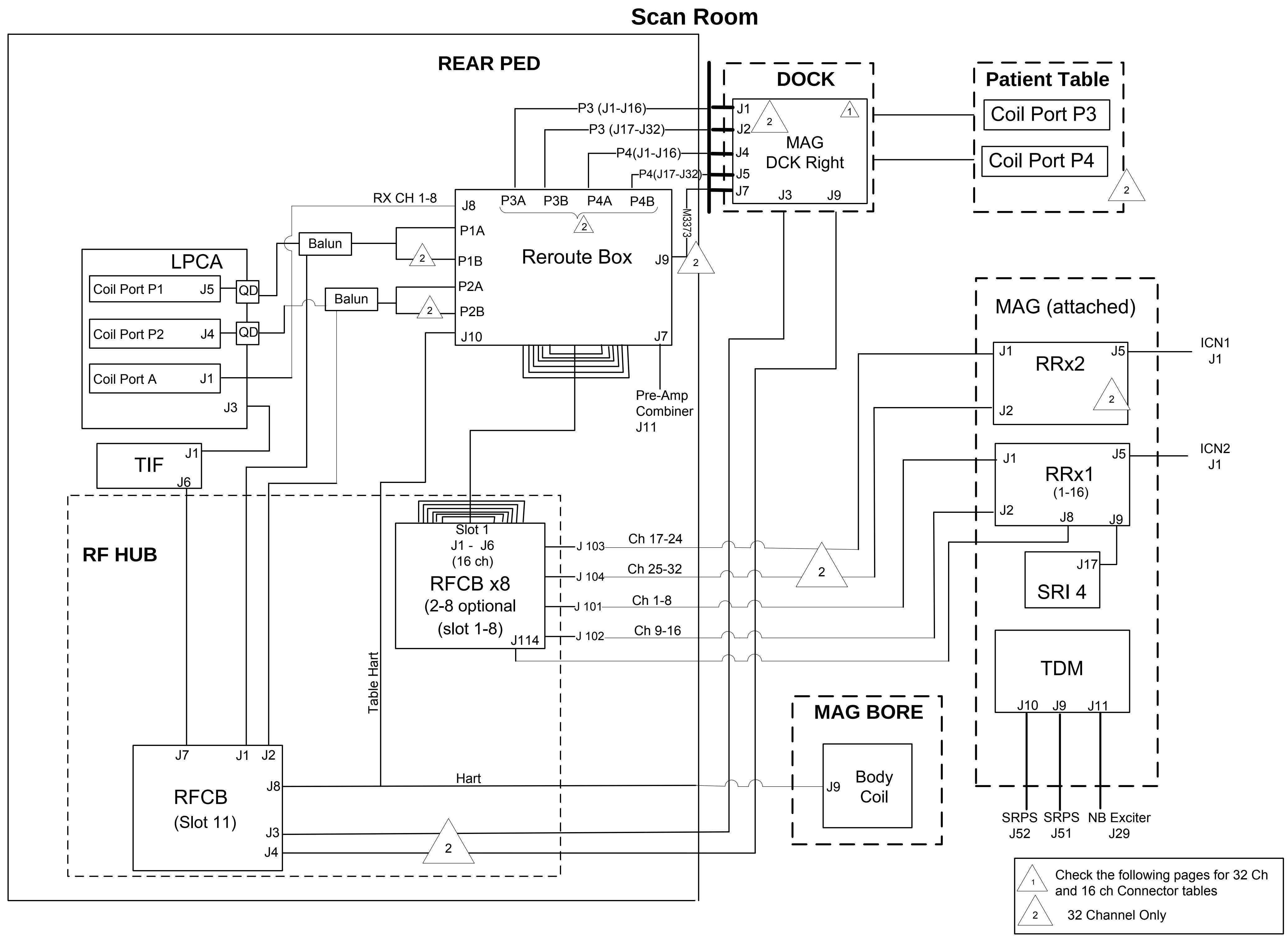


SIGNALS

1 BODY HYBRID REFLECTED	4 BODY RF IN, 35 kW PEP (75.44 dBm) nominal
2 RF REFLECTED POWER	5 Direct Drive BIAS INPUT
3 RECEIVE OUT	6 RECEIVE OUT, +15VDC Supply

CAM/RF
16 & 32 CHANNEL BODY TRANSMIT & RECEIVE BLOCK DIAGRAM



CAM/RF

Receive Chain - Reroute Box for 16 Channel and 32 Channel

Table 2

J8 A-CONN AND J7 BODY					
CONNECTOR	ONE END POSITION	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
J7 Body	1	N/A	Slot1 - J4	9	FCT A1096
J8 A-CONN	1	FCT FMX005102	Slot1 - J3	1	
J8 A-CONN	2		Slot1 - J3	2	
J8 A-CONN	3		Slot1 - J3	3	
J8 A-CONN	4		Slot1 - J3	4	
J8 A-CONN	5		Slot1 - J3	5	
J8 A-CONN	6		Slot1 - J3	6	
J8 A-CONN	7		Slot1 - J3	7	
J8 A-CONN	8	FCT FMX005102	Slot1 - J3	8	FCT A1096

CONNECTOR P3 - A					
CONNECTOR	ONE END POSITION	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P3 - A	1	FCT A1096	S1 - J3	9	FCT A1096
P3 - A	2		S1 - J3	10	
P3 - A	3		S1 - J3	11	
P3 - A	4		S1 - J3	12	
P3 - A	5		S1 - J3	9	
P3 - A	6		S1 - J3	10	
P3 - A	7		S1 - J3	11	
P3 - A	8		S1 - J3	12	
P3 - A	9		S1 - J4	1	
P3 - A	10		S1 - J4	2	
P3 - A	11		S1 - J4	3	
P3 - A	12		S1 - J4	4	
P3 - A	13		S1 - J4	1	
P3 - A	14		S1 - J4	2	
P3 - A	15		S1 - J4	3	
P3 - A	16	FCT A1096		4	FCT A1096

Connector P1-A					
CONNECTOR	ONE END POSITON	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P1-A	1	FCT A1096	S1 - J6	9	FCT A1096
P1-A	2		S1 - J6	10	
P1-A	3		S1 - J6	11	
P1-A	4		S1 - J6	12	
P1-A	5		S1 - J6	9	
P1-A	6		S1 - J6	10	
P1-A	7		S1 - J6	11	
P1-A	8		S1 - J6	12	
P1-A	9		S1 - J2	5	
P1-A	10		S1 - J2	6	
P1-A	11		S1 - J2	7	
P1-A	12		S1 - J2	8	
P1-A	13		S1 - J2	5	
P1-A	14		S1 - J2	6	
P1-A	15		S1 - J2	7	
P1-A	16	FCT A1096	S1 - J2	8	FCT A1096

CONNECTOR P3 - B					
CONNECTOR	ONE END POSITION	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P3 - B	1	FCT A1096	S1 - J4	5	FCT A1096
P3 - B	2		S1 - J4	6	
P3 - B	3		S1 - J4	7	
P3 - B	4		S1 - J4	8	
P3 - B	5		S1 - J4	5	
P3 - B	6		S1 - J4	6	
P3 - B	7		S1 - J4	7	
P3 - B	8		S1 - J4	8	
P3 - B	9		S1 - J3	13	
P3 - B	10		S1 - J3	14	
P3 - B	11		S1 - J3	15	
P3 - B	12		S1 - J3	16	
P3 - B	13		S1 - J3	13	
P3 - B	14		S1 - J3	14	
P3 - B	15		S1 - J3	15	
P3 - B	16	FCT A1096	S1 - J3	16	FCT A1096

Connector P1 - B					
CONNECTOR	ONE END POSITON	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P1 - B	1	FCT A1096	S1 - J5	1	FCT A1096
P1 - B	2		S1 - J5	2	
P1 - B	3		S1 - J5	3	
P1 - B	4		S1 - J5	4	
P1 - B	5		S1 - J5	1	
P1 - B	6		S1 - J5	2	
P1 - B	7		S1 - J5	3	
P1 - B	8		S1 - J5	4	
P1 - B	9		S1 - J1	13	
P1 - B	10		S1 - J1	14	
P1 - B	11		S1 - J1	15	
P1 - B	12		S1 - J1	16	
P1 - B	13		S1 - J1	13	
P1 - B	14		S1 - J1	14	
P1 - B	15		S1 - J1	15	
P1 - B	16	FCT A1096	S1 - J1	16	FCT A1096

Connector P4 - A					
CONNECTOR	ONE END POSITON	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P4 - A	1	FCT A1096	S1 - J5	9	FCT A1096
P4 - A	2		S1 - J5	10	
P4 - A	3		S1 - J5	11	
P4 - A	4		S1 - J5	12	
P4 - A	5		S1 - J5	9	
P4 - A	6		S1 - J5	10	
P4 - A	7		S1 - J5	11	
P4 - A	8		S1 - J5	12	
P4 - A	9		S1 - J1	5	
P4 - A	10		S1 - J1	6	
P4 - A	11		S1 - J1	7	
P4 - A	12		S1 - J1	8	
P4 - A	13		S1 - J1	5	
P4 - A	14		S1 - J1	6	
P4 - A	15		S1 - J1	7	
P4 - A	16	FCT A1096	S1 - J1	8	FCT A1096

Connector P2 - A					
CONNECTOR	ONE END POSITON	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P2 - A	1	FCT A1096	S1 - J2	1	FCT A1096
P2 - A	2		S1 - J2	2	
P2 - A	3		S1 - J2	3	
P2 - A	4		S1 - J2	4	
P2 - A	5		S1 - J2	1	
P2 - A	6		S1 - J2	2	
P2 - A	7		S1 - J2	3	
P2 - A	8		S1 - J2	4	
P2 - A	9		S1 - J2	5	
P2 - A	10		S1 - J2	6	
P2 - A	11		S1 - J2	7	
P2 - A	12		S1 - J2	8	
P2 - A	13		S1 - J2	5	
P2 - A	14		S1 - J2	6	
P2 - A	15		S1 - J2	7	
P2 - A	16	FCT A1096	S1 - J2	8	FCT A1096

Connector P4 - B					
CONNECTOR	ONE END POSITON	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P4 - B	1	FCT A1096	S1 - J6	1	FCT A1096
P4 - B	2		S1 - J6	2	
P4 - B	3		S1 - J6	3	
P4 - B	4		S1 - J6	4	
P4 - B	5		S1 - J6	1	
P4 - B	6		S1 - J6	2	
P4 - B	7		S1 - J6	3	
P4 - B	8		S1 - J6	4	
P4 - B	9		S1 - J6	13	
P4 - B	10		S1 - J6	14	
P4 - B	11		S1 - J6	15	
P4 - B	12		S1 - J6	16	
P4 - B	13		S1 - J6	13	
P4 - B	14		S1 - J6	14	
P4 - B	15		S1 - J6	15	
P4 - B	16	FCT A1096	S1 - J6	16	FCT A1096

Connector P2 - B					
CONNECTOR	ONE END POSITON	TERMINAL	CONNECTOR	OTHER END POSITION	TERMINAL
P2 - B	1	FCT A1096	S1 - J1	9	FCT A1096
P2 - B	2		S1 - J1	10	
P2 - B	3		S1 - J1	11	
P2 - B	4		S1 - J1	12	
P2 - B	5		S1 - J1	9	
P2 - B	6		S1 - J1	10	
P2 - B	7		S1 - J1	11	
P2 - B	8		S1 - J1	12	
P2 - B	9		S1 - J5	13	
P2 - B	10		S1 - J5	14	
P2 - B	11		S1 - J5	15	
P2 - B	12		S1 - J5	16	
P2 - B	13		S1 - J5	13	
P2 - B	14		S1 - J5	14	
P2 - B	15		S1 - J5	15	
P2 - B	16	FCT A1096	S1 - J5	16	FCT A1096

Table 1		
CONNECTOR	LABEL MARK	LABEL COLOR
S1 - J1	S1 J1	Blue
S1 - J2	S1 J2	
S1 - J3	S1 J3	
S1 - J4	S1 J4	
S1 - J5	S1 J5	
S1 - J6	S1 J6	
S1 - J1	S1 J1	Yellow
S1 - J2	S1 J2	
S1 - J3	S1 J3	
S1 - J4	S1 J4	
S1 - J5	S1 J5	
S1 - J6	S1 J6	

CAM/RF

Receive Chain - Reroute Box for 16 Channel and 32 Channel

Table 2

J8 - A CONN AND J7 BODY					
CONNECTOR	ONE END POSITION	TERMINAL	CONNECTOR	ONE END POSITION	TERMINAL
J7 Body	1	N/A	S1 - J4	9	FCT A1096
J8 - A CONN	1	FCT FMX005P102	S1 - J3	1	
J8 - A CONN	2		S1 - J3	2	
J8 - A CONN	3		S1 - J3	3	
J8 - A CONN	4		S1 - J3	4	
J8 - A CONN	5		S1 - J3	5	
J8 - A CONN	6		S1 - J3	6	
J8 - A CONN	7		S1 - J3	7	
J8 - A CONN	8	FCT FMX005P102	S1 - J3	8	FCT A1096

CONNECTOR P1 - A					
CONNECTOR	ONE END POSITION	TERMINAL	CONNECTOR	ONE END POSITION	TERMINAL
P1 - A	1	FCT A1096	S1 - J6	9	FCT A1096
P1 - A	2		S1 - J6	10	
P1 - A	3		S1 - J6	11	
P1 - A	4		S1 - J6	12	
P1 - A	5		S1 - J2	5	
P1 - A	6		S1 - J2	6	
P1 - A	7		S1 - J2	7	
P1 - A	8		S1 - J2	8	
P1 - A	9		S1 - J5	1	
P1 - A	10		S1 - J5	2	
P1 - A	11		S1 - J5	3	
P1 - A	12		S1 - J5	4	
P1 - A	13		S1 - J1	13	
P1 - A	14		S1 - J1	14	
P1 - A	15		S1 - J1	15	
P1 - A	16	FCT A1096	S1 - J1	16	FCT A1096

CONNECTOR P2 - A					
CONNECTOR	ONE END POSITION	TERMINAL	CONNECTOR	ONE END POSITION	TERMINAL
P2 - A	1	FCT A1096	S1 - J2	1	FCT A1096
P2 - A	2		S1 - J2	2	
P2 - A	3		S1 - J2	3	
P2 - A	4		S1 - J2	4	
P2 - A	5		S1 - J6	5	
P2 - A	6		S1 - J6	6	
P2 - A	7		S1 - J6	7	
P2 - A	8		S1 - J6	8	
P2 - A	9		S1 - J1	9	
P2 - A	10		S1 - J1	10	
P2 - A	11		S1 - J1	11	
P2 - A	12		S1 - J1	12	
P2 - A	13		S1 - J5	13	
P2 - A	14		S1 - J5	14	
P2 - A	15		S1 - J5	15	
P2 - A	16	FCT A1096	S1 - J5	16	FCT A1096

Table 1		
Connector	Label Mark	Label Color
S1 - J1	S1 J1	Blue
S1 - J2	S1 J2	
S1 - J3	S1 J3	
S1 - J4	S1 J4	
S1 - J5	S1 J5	
S1 - J6	S1 J6	

CAM/RF
Receive Chain - Reroute Box for 16 Channel and 32 Channel

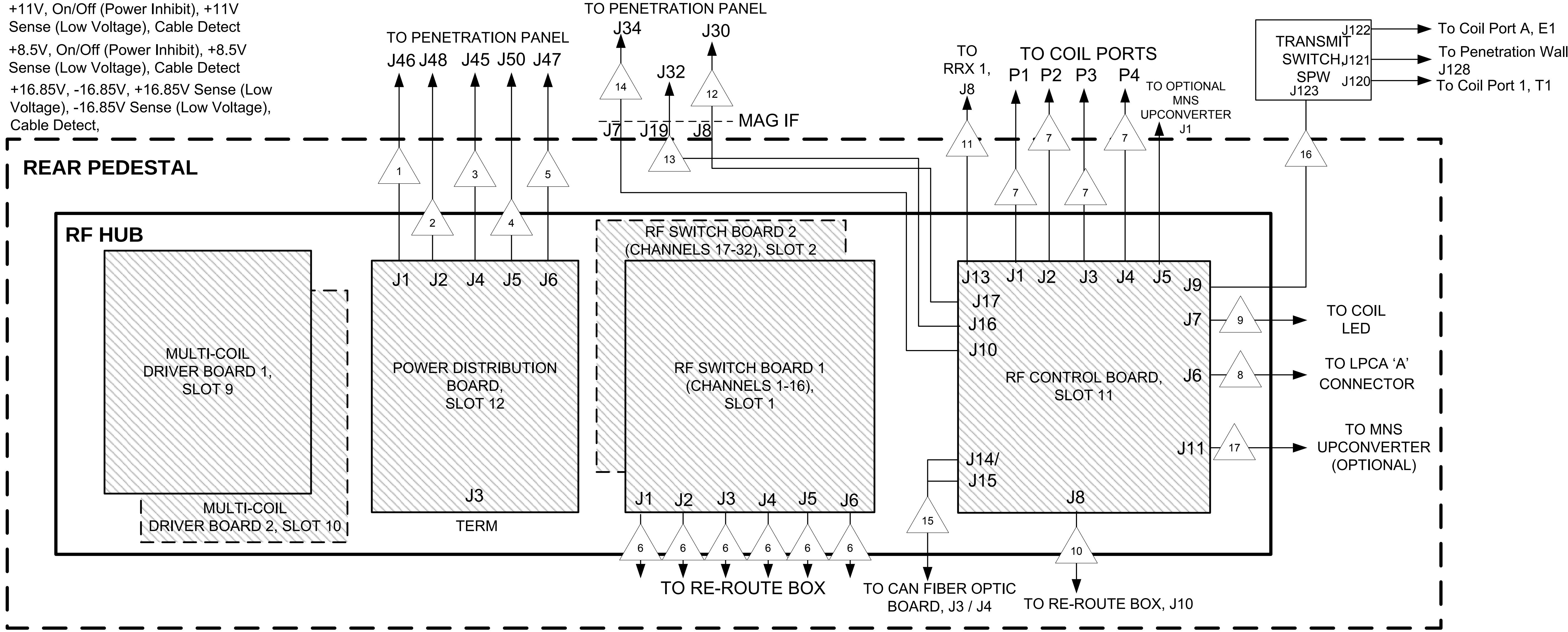
RF HUB Block Diagram

SIGNALS

1 — +11V, On/Off (Power Inhibit), +11V Sense (Low Voltage), Cable Detect

2 — +8.5V, On/Off (Power Inhibit), +8.5V Sense (Low Voltage), Cable Detect

3 — +16.85V, -16.85V, +16.85V Sense (Low Voltage), -16.85V Sense (Low Voltage), Cable Detect,



4 — -8.5V, On/Off (Power Inhibit), -8.5V Sense (Low Voltage), Cable Detect

5 — -11V, On/Off (Power Inhibit), -11V Sense (Low Voltage), Cable Detect

6 — 16MHz RF IF data, 16 Channels, LO, Diagnostic, Control, and MC Bias Signals

7 — +10V DC, Coil Present, Body Transmit Enable, Coil ID, MUX Select Bits 0-4, Cable Connect, Ground

8 — +15V DC, +10V DC, -15V DC, Coil Present, Coil ID, Body Transmit Enable, MUX Select Bits 0-3, MC Bias Pins 0-7, Ground

9 — +5V DC, Green LED Port A, Red LED Port A, Auxiliary 1-8, Cable Connect, Ground

10 — +5V DC, Table ID, Cable Connect, 1-WIRE RRB, 1-WIRE TABLE, Ground

11 — Fiber Optic Transmitter, Strobe

12 — Twinax, Low Oscillator (112 MHz)

13 — Coax, Loopback (127.72 MHz)

14 — Narrowband Unblank, Broadband Unblank, Inhibit Unblank, Continuous Wave Unblank

15 — Fiber Optic, Diagnostic, Control Communication

16 — +15V DC, Bits 1-3 of Transmit Switch Control, Cable Detect Tx Switch, Ground

17 — +10V DC, Cable Connect, 1-Wire ID, Switch Control Bits 0-1, Loopback Control Bits 0-1, LO Detect, Power Good, MNS LO CBL DET, MNS LB CBL DET, Auxiliary (0,1,5), Ground

Multi-Coil Driver Board (MCD)

INDICATOR LEDS:
Active When Lit:
Enable (1-8)
7V MODE (1-8)
10V MODE (1-8)

POWER GOOD LEDS:

LED	DESCRIPTION	LED DARK	LED ACTIVE
+/- 10V'	Power Drivers, Analog Devices	Deselected	Selected
+/- 7V'	Power Drivers, Analog Devices	Deselected	Selected
+/- 5V'	Digital Power	Fault	Normal
+1.2V'	FPGA CORE Volts	Fault	Normal
+3.3V'	FPGA I/O Volts	Fault	Normal

FAULT LEDS:

Fault LED	Description	LED Active
TEMP	Board Temperature	Fault
BIAS	MC Bias	Fault

Radio Frequency Control Board (RFCB)

LED DIAGNOSTICS:

Signal/LED Name	Color	Description
CAN_STAT	Green	CAN status
CAN_ERROR	Green	CAN error
+1V2 Digital Power'	Green	+1V2 Digital Power'
+3V3 Digital Power Good'	Green	+3V3 Digital Power Good'
+5V Digital Power Good'	Green	+5V Digital Power Good'
+15V Digital Power Good'	Green	+15V Digital Power Good'
-15V Digital Power Good'	Green	-15V Digital Power Good'
+10V Digital Power Good'	Green	+10V Digital Power Good'
-10V Digital Power Good'	Green	-10V Digital Power Good'
LB_GOOD	Green	Loopback Good
LO_GOOD	Green	Local Oscillator Good
UNBLANK	Green	Unblank signal
INHIBIT	Yellow	Inhibit signal
-5V Digital Power Good'	Green	-5V Digital Power Good'

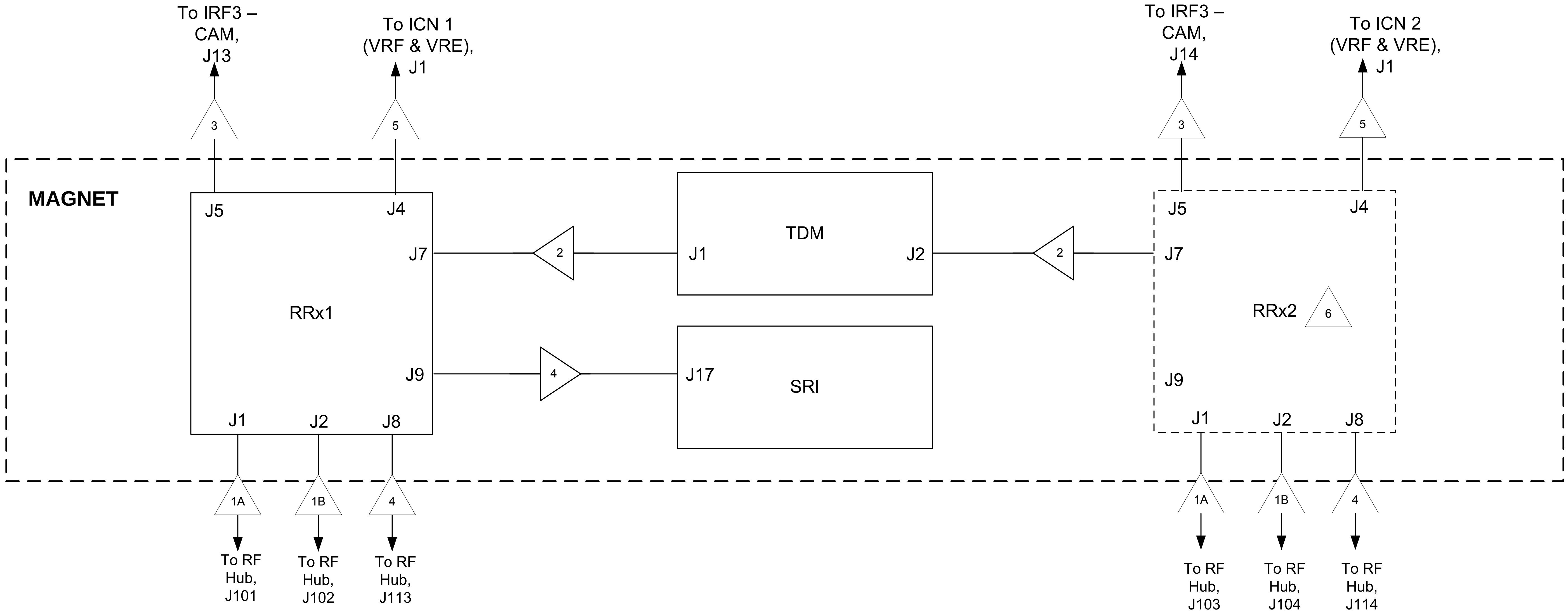
Power Distribution Board (PDB)

LED	Purpose	Color
+16V'	+16.85V Power Good'	Green
-16V'	-16.85V Power Good'	Green
+11V'	+11.6V Power Good'	Green
-11V'	-11.6V Power Good'	Green
+8V'	+8.5V Power Good'	Green
-8V'	-8.5V Power Good'	Green
+3.3V'	+3.3 DIG Power Regultated	Green
+5V'	+3.3 DIG RFCB Power Regultated	Green

Radio Frequency Switch Board (RFSB)

POWER SUPPLY LEDS:

LED	Description
+5D'	+5D Power Monitoring'
+5V'	5V Power Monitoring
+6V'	+6V Power Monitoring'
-5V'	-5V Power Monitoring'



SIGNALS

1A — +3V3 DC from RF Hub, RF Hub Ground, Cable Connection Confidence Loop, Port IDs 0-2, Power Supply Good for RFCB, CAN Hard Reset for RRx, CAN Rx Link Active on RFCB, RFCB Clock Enable, IRF Link Status, VRF Link Status, Reset CPLD on RFCB, CFB and RFCB Loopback, Channels 1 -7 (Receive)

1B — +3V3 DC from RF Hub, RF Hub Ground, Cable Connection Confidence Loop, Port IDs 0-2, Power Supply Good for RFCB, CAN Hard Reset for RRx, CAN Rx Link Active on RFCB, RFCB Clock Enable, IRF Link Status, VRF Link Status, Reset CPLD on RFCB, CFB and RFCB Loopback, Channels 8 -16 (Receive)

2 — +6.85V Digital Power, +6.85V Analog Power, -6.85V Analog Power, +3.16V Digital Power, +5.05V Digital Power, +5.05V Analog Power, Cable Connect Confidence Loop, Ground, TNS Spike Detect P, TNS Spike Detect N, TNS Test Inject Trigger, TNS Test Inject Amplitude, TNS Threshold, TDM Test Mode Enable, 80MHz Clock Input P, 80MHz Clock Input N, Ground

3 — Fiber Optic, DVMR Link

4 — Fiber Optic, STRB (Strobe)

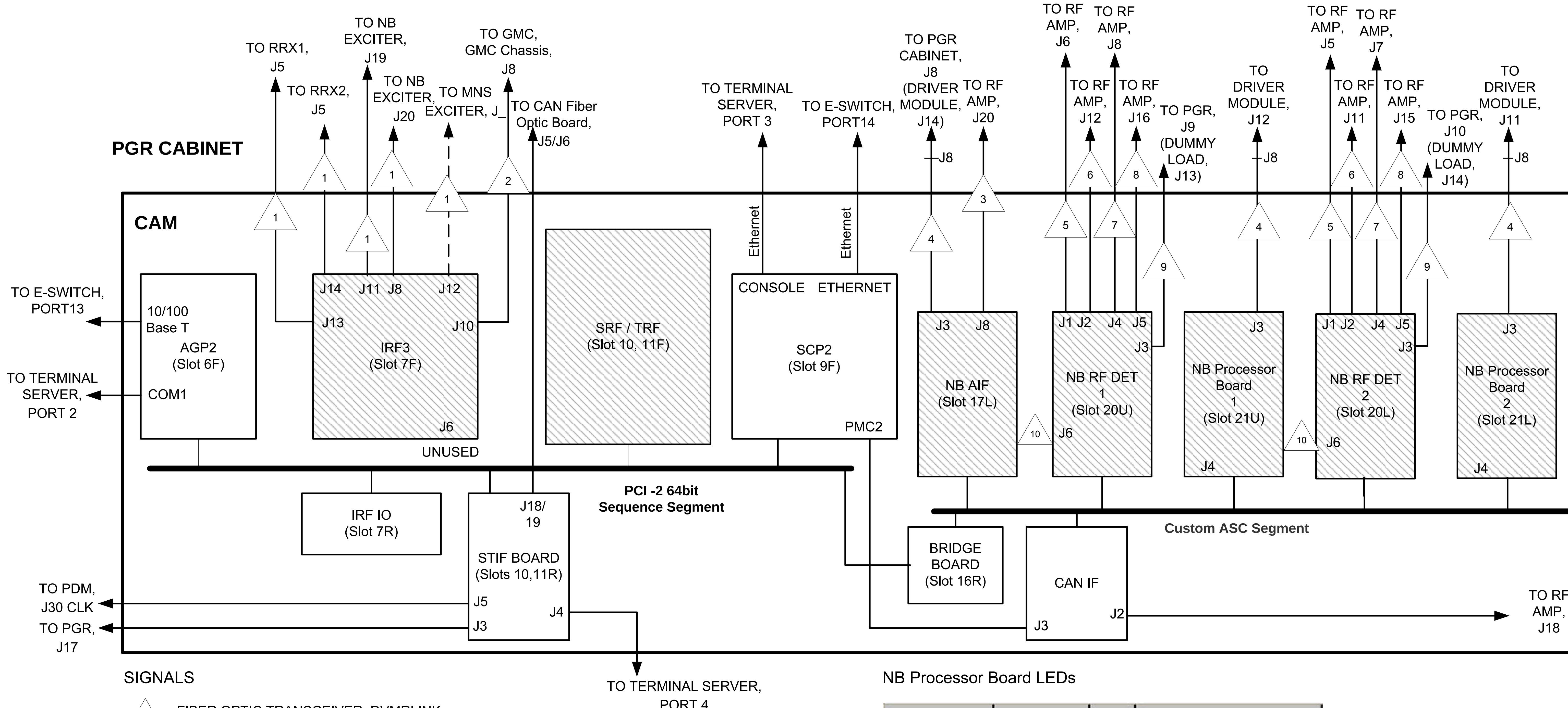
5 — Fiber Optic, VRF Link

6 — RRX2 present in 32-Channel only

RRx LED DIAGNOSTICS

Reference Designator	Signal or LED Name	Color	Externally Visible	Description
DS1	HUB_PWR_GOOD	Green	Yes	Lit = RF Hub +3.3V power supply good for connector J1
DS2	PORT_ID_2	Green	Yes	See Table 1 for Port ID decoding.
DS3	PORT_ID_1	Green	Yes	See Table 1 for Port ID decoding.
DS4	PORT_ID_0	Green	Yes	See Table 1 for Port ID decoding.
DS5	CLOCK_ACTIVE	Green	Yes	Lit = 80MHz master clock active
DS6	POWER_GOOD	Green	Yes	Lit = RRx power is good
DS7	IRF_LINK_UP	Green	Yes	Lit = IRF link up, Blinking = IRF link error
DS8	VRF_LINK_UP	Green	Yes	Lit = VRF link up, Blinking = VRF link error

CAM/RF
RRX Module Block Diagram



SIGNALS

1 FIBER OPTIC TRANSCEIVER, DVMRLINK SEQUENCE BUS DATA

2 Fiber Optic, 80 MHz Clock

3 RS422, UNBLANK Input to AMP from ASC, CAN Power OK, Amp Power Supply to CIB OK, CABLE DETECT, System RESET input to AMP from ASC

4 RS422, UNBLANK, INHIBIT

5 FWD_CH1, 8MHz to 300MHz

6 REFL_CH1, 8MHz to 300MHz

7 FWD_CH2, 8MHz to 300MHz

8 REFL_CH2, 8MHz to 300MHz

9 BODY HYBRID REFLECTED

10 50 Ohm Terminator

NB Processor Board LEDs

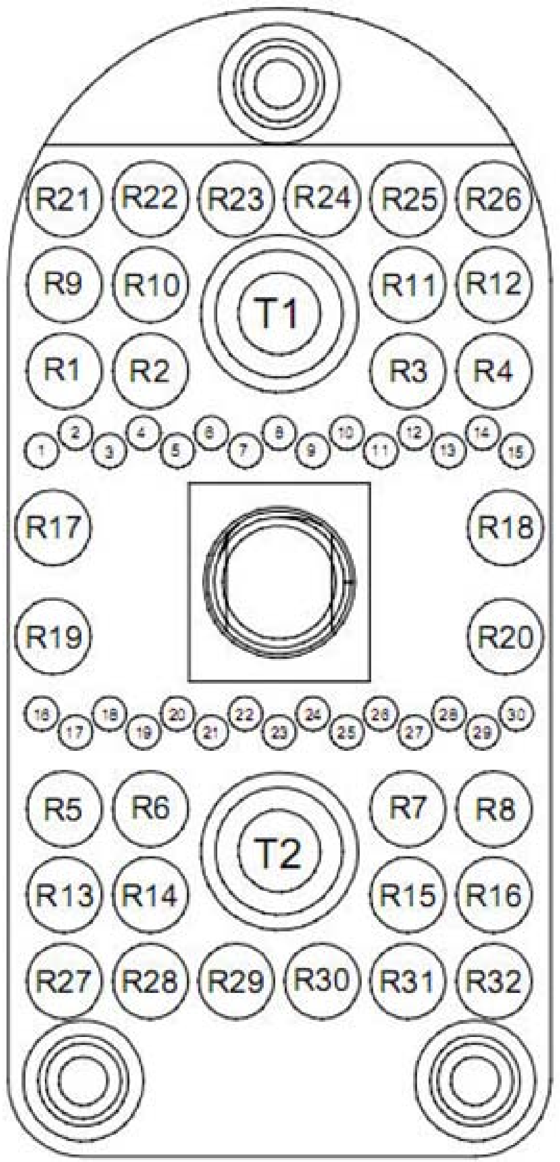
LED Silkscreen Name	Signal Name	Color	Description
1V5	1V5 Power OK	Green	Lit = +1.5DC power present
CN_PVR	CAN POVER OK	Green	Lit = Isolated +24V for CAN present
CN_STAT	CAN_STAT	Green	Flashes
INHBT	INHBT_UNBLK_N	Yellow	Lit = Power Monitored error condition, Unblank is Inhibited

SRF / TRF BOARD LEDs

Display	Color	Description
1.8V	Green	1.8V Power Ready
2.5V	Green	2.5V Power Ready
DSP0	Green	Sequence DSP0 Active
DSP1	Green	Sequence DSP1 Active
1.9V	Green	1.9V Power Ready
WARP	Green	WARP Alive
SPU	Green	SPU DUART Ch, A MF

IRF3 BOARD LEDs

Mnemonic	Function
J10 RDY	Link-1 Status
J11 RDY	Link-2 Status
J12 RDY	Link-3 Status
J13 RDY	Link-4 Status
J14 RDY	Link-5 Status
REF OK	Reference Clock Detected
PWR OK	Voltage OK
RST REQ	System Reset Request



P-port pin #	Signal Name	Description	Notes
1	Coil_Pres	Coil Present	Must Jumper pins 1 and 2 in the coil
2	Coil_Pres_Rtn	Coil Present Return	
3	Body_Tx_Enable	Body Transmit Enable	Jumper pins 3 and 4 on receive-only coils
4	Body_Tx_Enable_Rtn	Body Transmit Enable Return	
5	Coil_ID	Coil ID	Dallas 1-wire
6	Coil_ID_Rtn	Coil ID Return	
7	MUX_Select_0	MUX Select Bit 0	
8	MUX_Select_0_Rtn	MUX Select Bit 0 Return	
9	MUX_Select_1	MUX Select Bit 1	
10	MUX_Select_1_Rtn	MUX Select Bit 1 Return	
11	MUX_Select_2	MUX Select Bit 2	
12	MUX_Select_2_Rtn	MUX Select Bit 2 Return	
13	MUX_Select_3	MUX Select Bit 3	
14	MUX_Select_3_Rtn	MUX Select Bit 3 Return	
15	n/c	Reserved	
16	MUX_Select_4	MUX Select Bit 4	
17	MUX_Select_4_Rtn	MUX Select Bit 4 Return	
18	+10V_DC	+10V_DC	Up to 3 amp continuous aux. Power is available at each P-port.
19	AUX_PWR_Rtn	Auxiliary Power Return	
20	+10V_DC	+10V_DC	
21	AUX_PWR_Rtn	Auxiliary Power Return	
22	+10V_DC	+10V_DC	
23	AUX_PWR_Rtn	Auxiliary Power Return	
24	n/c	Reserved	
25	n/c	Reserved	
26	n/c	Reserved	
27	n/c	Reserved	
28	n/c	Reserved	
29	n/c	Reserved	
30	n/c	Reserved	

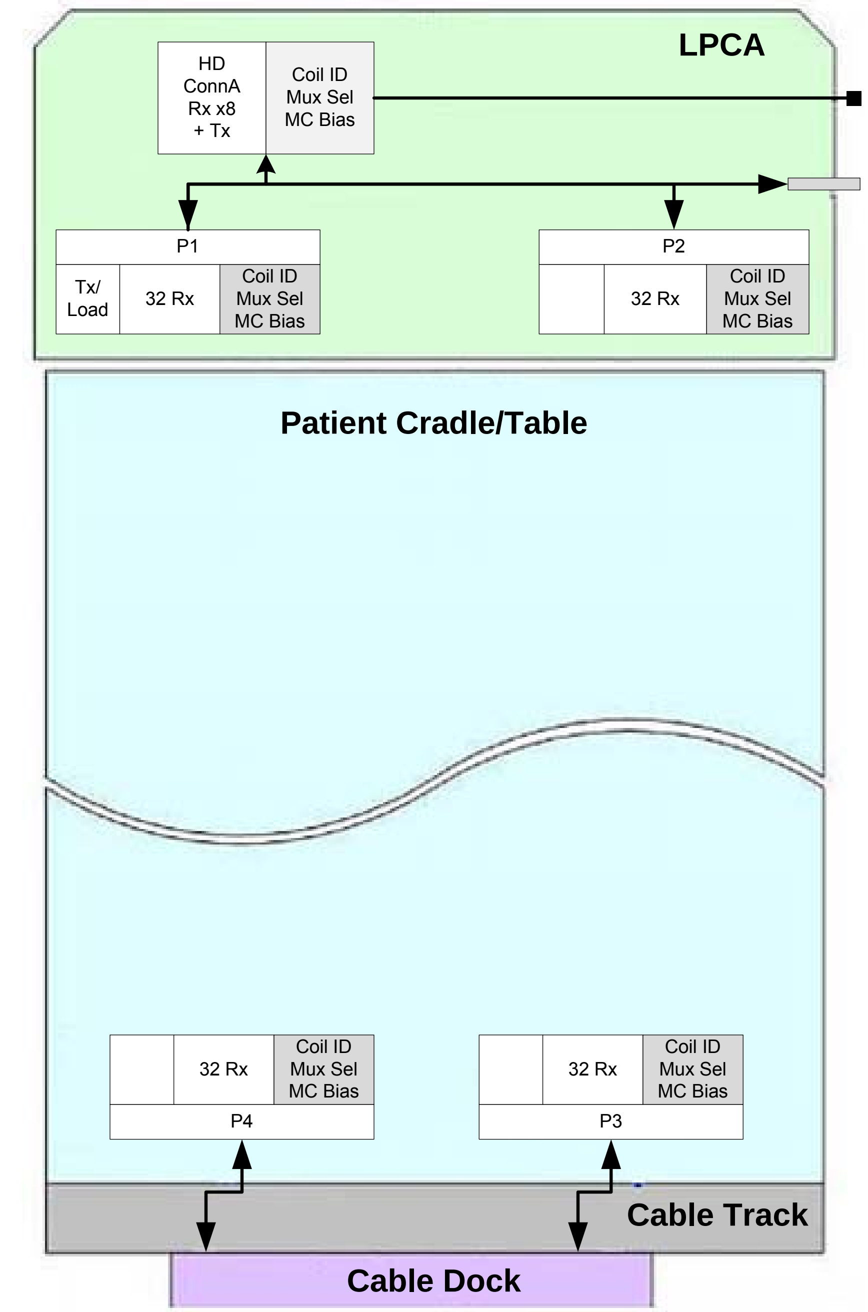
TRANSMIT (TX) PIN ASSIGNMENTS

P-port pin #	Signal Name	Description	Notes
T1	Tx	Transmit	These available only on A1 Connector.
T2	Reflected pwr	Reflected power	

RECEIVE (RX) PIN ASSIGNMENTS

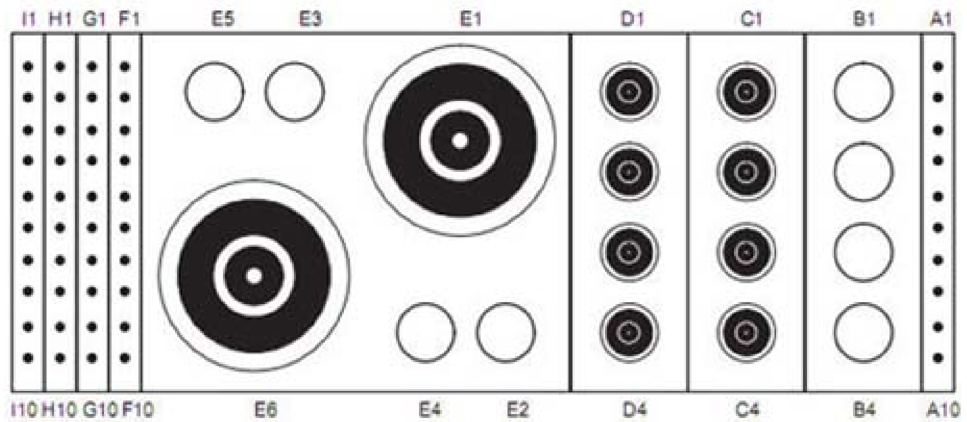
P-port pin #	Signal Name	Description	Notes
R1-R32	Rx	Receive and MC Bias	Receive Signal and MC Bias

See sheet PH 3-1 for more details on the patient handling table connections.



CAM/RF
'P' Coil Connectors

View from CONNECTOR SIDE of Connector Assembly

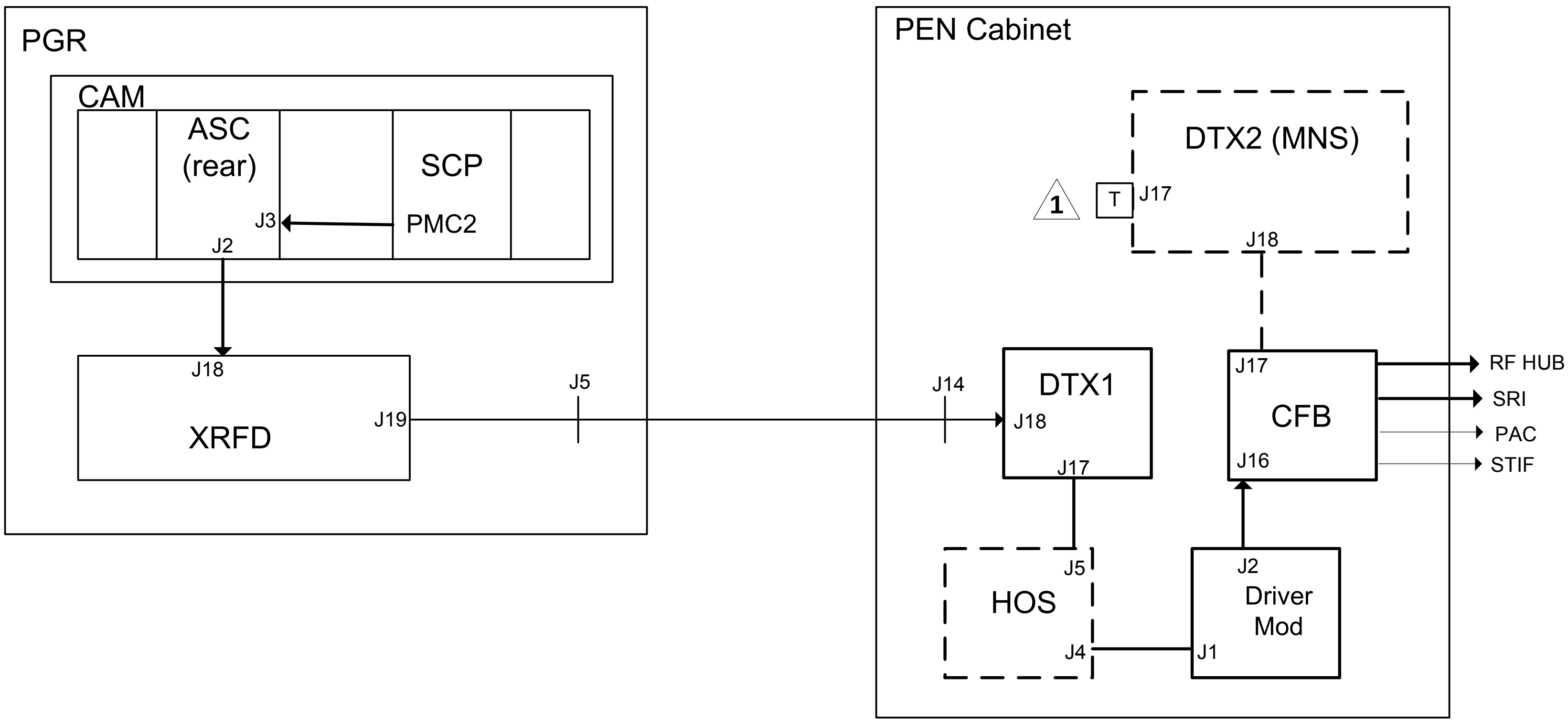


Contact Function Allocation

Pin	Twisted with	Description	Comments
A1	A2	coil present	
A2	A1	coil present return	
A3	A4	multiplexerSelect0	RS-422+
A4	A3	multiplexerSelect0 return	RS-422-
A5	A6	multiplexerSelect1	RS-422+
A6	A5	multiplexerSelect1 return	RS-422-
A7	A8	multiplexerSelect2	RS-422+
A8	A7	multiplexerSelect2 return	RS-422-
A9	A10	multiplexerSelect3	RS-422+
A10	A9	multiplexerSelect3 return	RS-422-
B1	n/a	Key blocked in receptacle E	
B2	n/a	Key blocked in receptacle F	
B3	n/a	Key blocked in receptacle G	
B4	n/a	Key blocked in receptacle H	
C1	n/a	MC-1A,B or MC-9B	Preamp out 1, no bias, DC blocked
C2	n/a	MC-2A,B or MC-10B	Preamp out 2, no bias, DC blocked
C3	n/a	MC-3A,B or MC-11B	Preamp out 3, no bias, DC blocked
C4	n/a	MC-4A,B or MC-12B	Preamp out 4, no bias, DC blocked
D1	n/a	MC-5A,B or MC-13B	Preamp out 5, no bias, DC blocked
D2	n/a	MC-6A,B or MC-14B	Preamp out 6, no bias, DC blocked
D3	n/a	MC-7A,B or MC-15B	Preamp out 7, no bias, DC blocked
D4	n/a	MC-8A,B or MC-16B	Preamp out 8, no bias, DC blocked

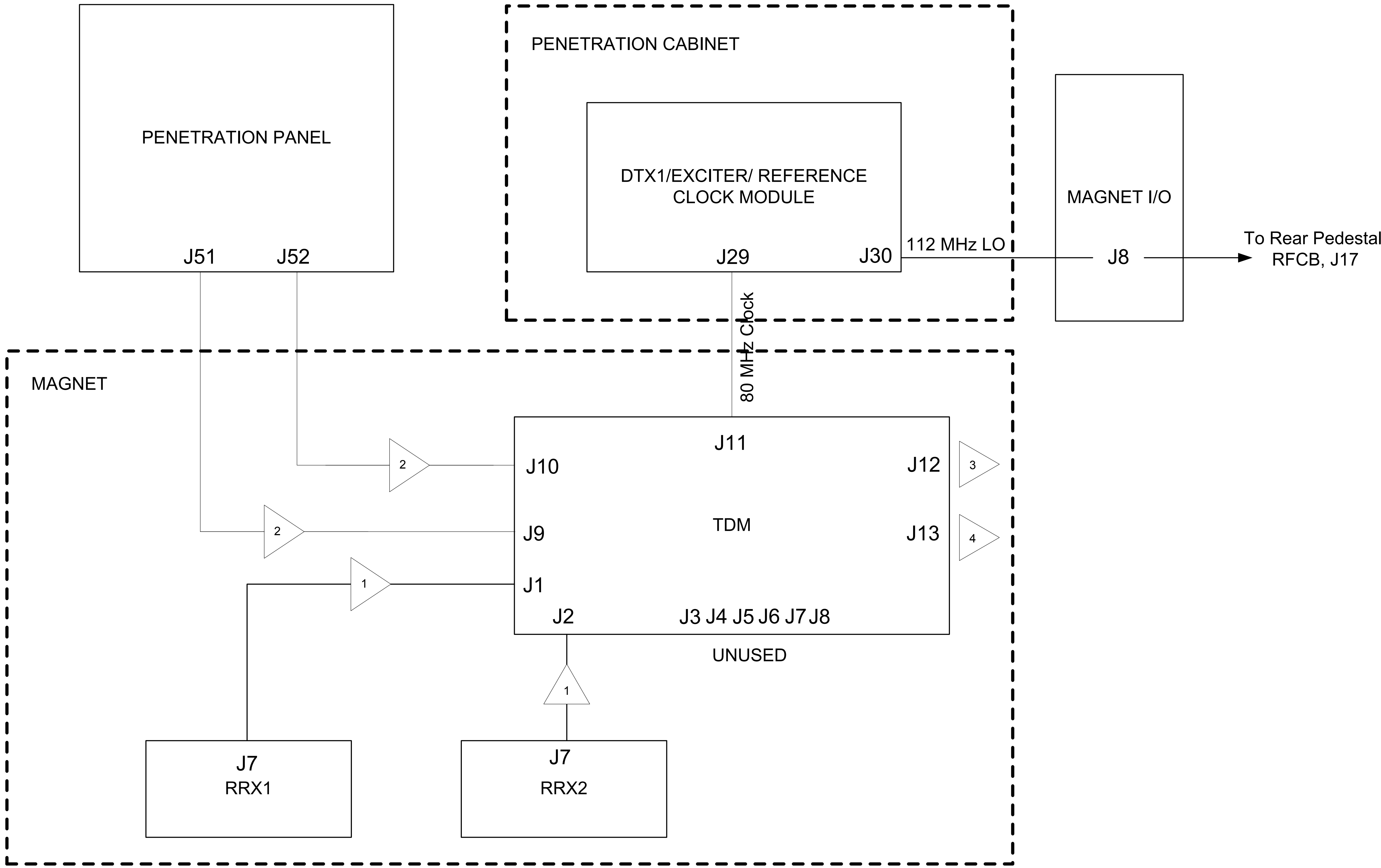
Contact Function Allocation

Pin	Twisted with	Description	Comments
E1	n/a	Transmit signal up to 2kw (1.5T) or 4kw (3.0T) PEP	"A" Connector only. Not present in "B" Connector
E2	n/a	Key blocked in receptacle C	
E3	n/a	Key blocked in receptacle A	Blocked for "A"
E4	n/a	Key blocked in receptacle B	Blocked for "B"
E5	n/a	Key blocked in receptacle D	
E6	n/a	Reflected power to 50 ohm load 2.5kw PEP, 50w RMS	"A" Connector only. Not present in "B" Connector
F1	G1	MC bias 1	MC T/R Bias Line 17 for "A", 25 for "B"
F2	G2	MC bias 2	MC T/R Bias Line 18 for "A", 26 for "B"
F3	G3	MC bias 3	MC T/R Bias Line 19 for "A", 27 for "B"
F4	G4	MC bias 4	MC T/R Bias Line 20 for "A", 28 for "B"
F5	G5	MC bias 5	MC T/R Bias Line 21 for "A", 29 for "B"
F6	G6	MC bias 6	MC T/R Bias Line 22 for "A", 30 for "B"
F7	G7	MC bias 7	MC T/R Bias Line 23 for "A", 31 for "B"
F8	G8	MC bias 8	MC T/R Bias Line 24 for "A", 32 for "B"
F9	F10	body transmit enable	
F10	F9	body transmit enable return	
G1	F1	MC bias return 1	MC T/R Bias Line 17 for "A", 25 for "B"
G2	F2	MC bias return 2	MC T/R Bias Line 18 for "A", 26 for "B"
G3	F3	MC bias return 3	MC T/R Bias Line 19 for "A", 27 for "B"
G4	F4	MC bias return 4	MC T/R Bias Line 20 for "A", 28 for "B"
G5	F5	MC bias return 5	MC T/R Bias Line 21 for "A", 29 for "B"
G6	F6	MC bias return 6	MC T/R Bias Line 22 for "A", 30 for "B"
G7	F7	MC bias return 7	MC T/R Bias Line 23 for "A", 31 for "B"
G8	F8	MC bias return 8	MC T/R Bias Line 24 for "A", 32 for "B"
G9	n/a	Reserved	
G10	n/a	Reserved	
H1	n/a	Reserved	
H2	n/a	Reserved	
H3	n/a	Reserved	
H4	n/a	Reserved	
H5	n/a	Reserved	
H6	n/a	Reserved	
H7	n/a	Reserved	
H8	n/a	Reserved	
H9	n/a	Reserved	
H10	n/a	Reserved	
I1	n/a	Reserved	
I2	I5	+15Vdc (auxiliary & preamp power)	
I3	I6	+10Vdc (auxiliary & preamp power)	
I4	n/a	-15Vdc (auxiliary power)	
I5	I2	Return for aux power	
I6	I3	Return for aux power	
I7	I8	coil ID (Dallas OneWire protocol)	All coils MUST have coil ID.
I8	I7	coil ID return (Dallas OneWire protocol)	All coils MUST have coil ID.
I9	I10	coil present'	
I10	I9	coil present return'	



1 TERMINATOR CONNECTED TO CFB J17 ON SYSTEMS WITHOUT MNS

CAM/RF
CAN Copper Link



1 +7VDC, +5VDC, +3VDC, -6VDC, 80 MHz Clock, Spike Detect Signal

2 +7VDC, +5VDC, +3VDC, -6VDC, Voltage Sense, 1-Wire Network

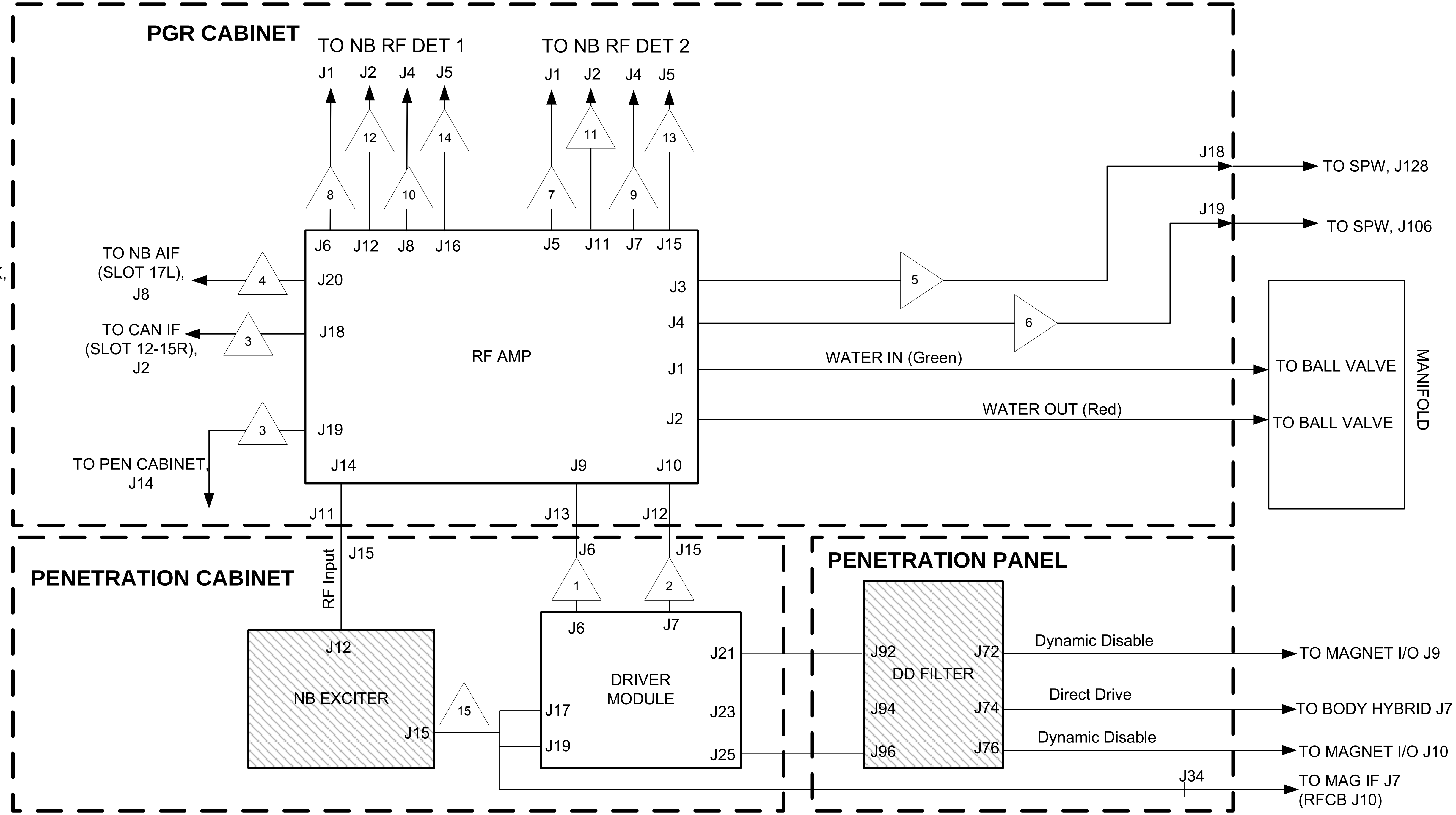
3 To TNS Loop Antenna

4 Transient Detection Port (Not Used)

CAM/RF
TDM

SIGNALS

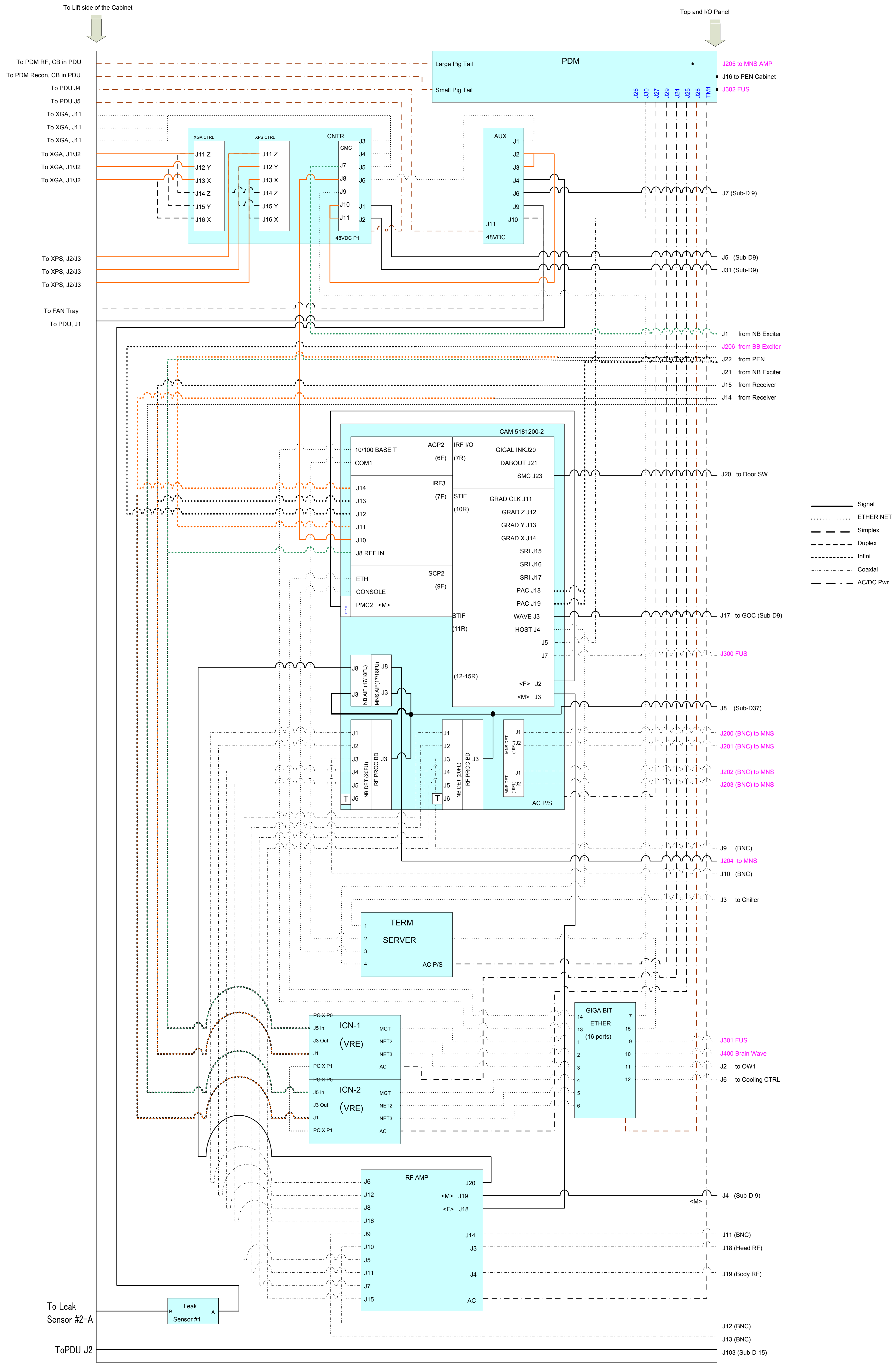
- 1 — Head TR Bias Input, +8V to -14V
- 2 — Body TR Bias Input, +4.5V to -14V
- 3 — CAN Low, CAN High, CAN Shield, Logic and 24V
- 4 — RS422, UNBLANK Input to AMP from ASC, CAN Power OK, Amp Power Supply to CIB OK, CABLE DETECT, CABLE DETECT RETURN, System RESET input to AMP from ASC
- 5 — HEAD RF OUT, 4.5 kW PEP (66.53 dBm) nominal
- 6 — BODY RF OUT, 35 kW PEP (75.44 dBm) nominal
- 7 — BODY FWD 1, -55 dB +/-0.5dB
- 8 — BODY FWD 2, -55 dB +/-0.5dB
- 9 — HEAD FWD 1, -45 dB +/-0.5dB
- 10 — HEAD FWD 2, -45 dB +/-0.5dB
- 11 — BODY REF 1, -50 dB +/-0.5dB
- 12 — BODY REF 2, -50 dB +/-0.5dB
- 13 — HEAD REF 1, -40 dB +/-0.5dB
- 14 — HEAD REF 2, -40 dB +/-0.5dB



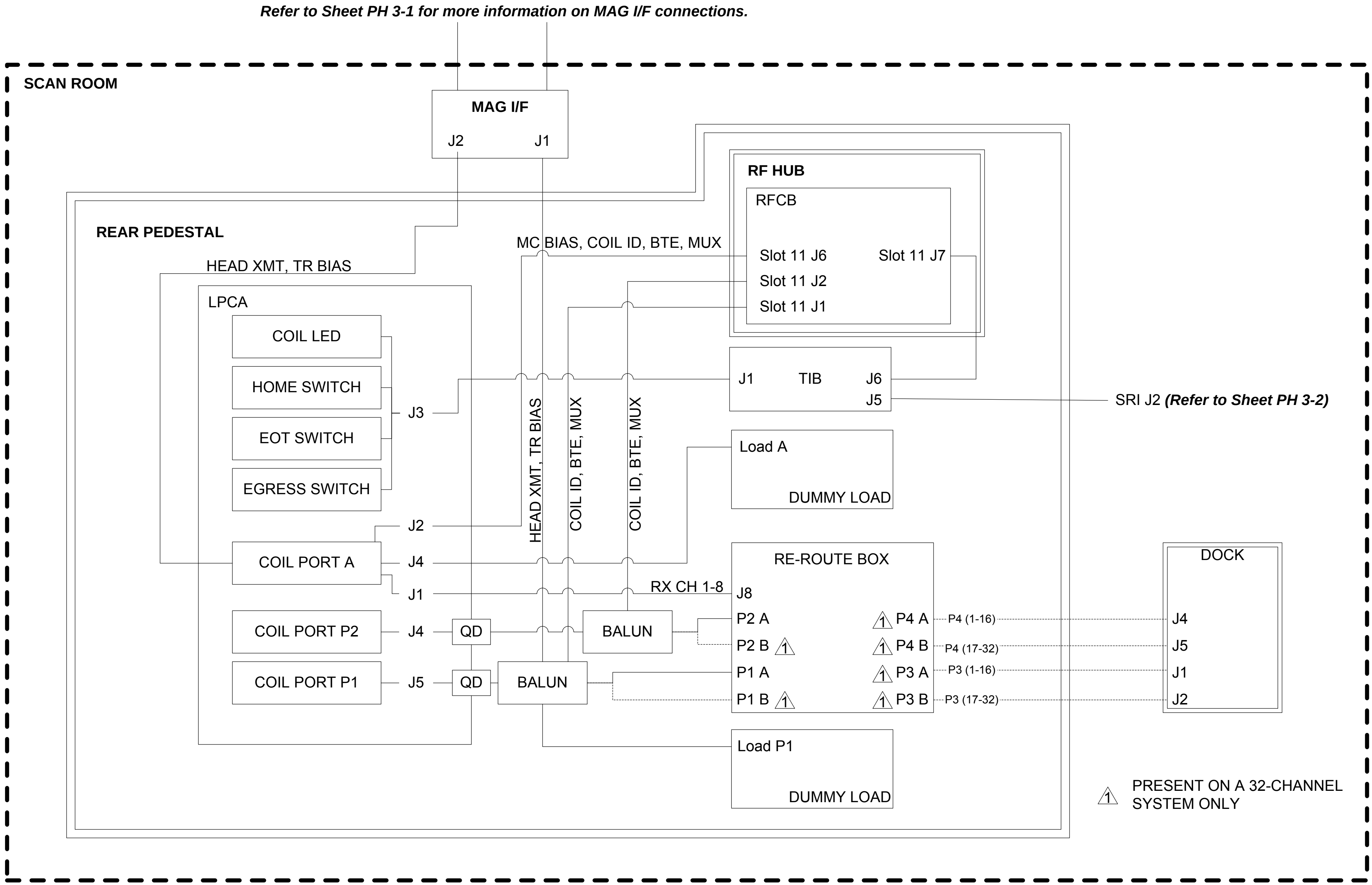
Status Condition	Definition	Front Panel Label & Display	Color
Fault state	Detection of an amplifier fault condition	Set FAULT indication	RED
RFLOCK signal asserted	System has asserted RFLOCK signal	Set RFLOCK indication	YELLOW
OPERATE state	Module in OPERATE state	Set OPERATE indication	GREEN
Wait state	Module in a transient state (processing a command)	Set WAIT indication	YELLOW
STANDBY state	Module in STANDBY state	Set STANDBY indication	GREEN
POWER OFF state	Module in POWER OFF state	Set POWER OFF indication	YELLOW
Head mode active	Module in Head mode	Set HEAD indication	GREEN
Body mode active	Module in Body mode	Set BODY indication	GREEN
Test mode active	Module in Test mode	Set TEST indication	GREEN
UNBLANK signal active	Presence of UNBLANK (See Note 1)	Toggle UNBLANK indication	YELLOW
HV Power Supply 1 charged	High voltage supply 1 for XRFD is active/not discharged to a safe level	Set DC HIGH VOLTAGE 1 indication	GREEN

15 Unblank

HV Power Supply 2 charged	High voltage supply 2 for XRFD is active/not discharged to a safe level	Set DC HIGH VOLTAGE 2 indication	GREEN
AC Power applied	AC line power to the amplifier module is ON	Set AC POWER indication	GREEN



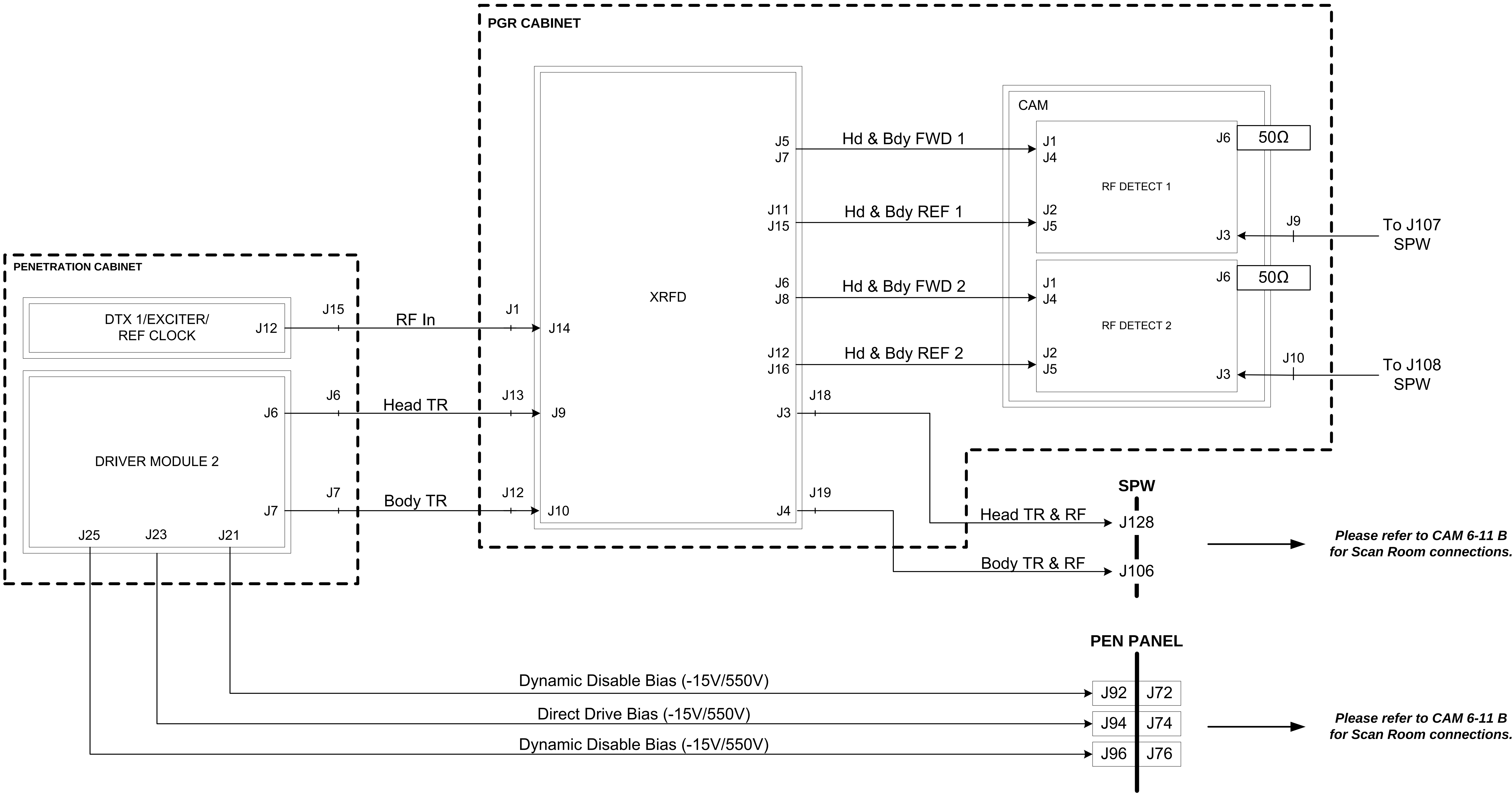
CAM/RF
Recon, RF, VRE SECTION



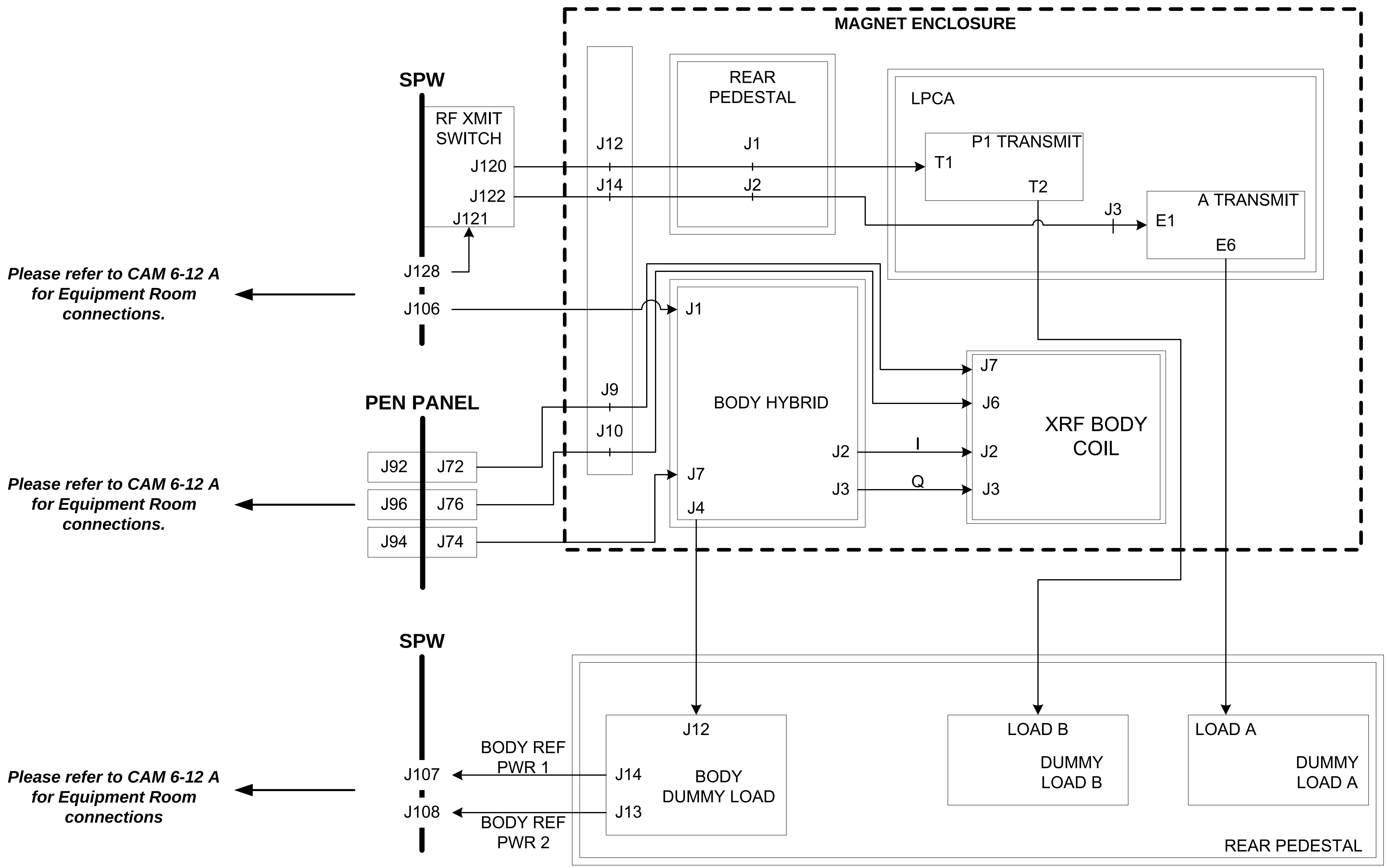
CAM/RF

LPCA INTERCONNECTS

MAGNET ROOM/SCAN ROOM – REAR PEDESTAL



CAM/RF
Transmit Chain Theory – Equipment Room



CAM/RF
Transmit Chain Theory – Scan Room